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6th Sem/ Mechanical Engineering
Subject : Automobile Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Automobile can be classified based on which of the following parameter?
- a) Fuel used b) Transmission
 - c) Drive d) All of the mentioned
- Q.2 Which of these were or are used in automobiles to provide suspension.
- a) Coil springs b) Torsion bars
 - c) Leaf springs d) All of the mentioned
- Q.3 What is the function of the alternator?
- a) Recharging the Battery
 - b) Voltage Regulator
 - c) Auto-Ignition
 - d) None of the above

- Q.4 Leaf spring absorbs shock by
a) Expanding b) Bending
c) Compression d) Twisting
- Q.5 What is the speed ratio of top shift position of gear box
a) 3.589 b) 2.166
c) 0.999 d) None of the above
- Q.6 In air brake air pressure is applied by
a) The engine manifold
b) A compressor
c) The diaphragm valve
d) A condenser

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Toe-Out. (CO3)
- Q.8 Name various passenger safety devices used in automobile vehicles. (CO3)
- Q.9 Expand ABS. (CO4)
- Q.10 ADAS stands for _____. (CO7)
- Q.11 What is regenerative braking system? (CO4)
- Q.12 What is rear wheel drive? (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is Automatic transmission system write its advantages in automobile. (CO2)
- Q.14 Explain the construction and working of Multi-Plate clutch. (CO2)
- Q.15 Explain wheel Balancing & Alignment.
- Q.16 Write the classification of Automobile. (CO1)
- Q.17 Differentiate between pure electric vehicle and hybrid vehicle. (CO7)
- Q.18 Write the explain the different types of rear axle. (CO2)
- Q.19 What are requirements of good braking system? (CO4)
- Q.20 Write the advantages of coil spring over leaf spring in suspension system. (CO3)
- Q.21 Write the various functions of battery in automobile. (CO5)
- Q.22 Explain Ackerman steering mechanism with the help of neat sketch. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the construction and working of vacuum brakes. (CO4)

Q.24 What is Dynamo. Give its principle, construction and working in detail. (CO7)

Q.25 Explain the construction and working of Sliding mesh gear box. (CO2)